Progress Report: Predicting Variant Pathogenicity in ClinVar

CMPS261 — Section L2 — Professor Rida Assaf — 6 November 2026
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1. Summary

This report covers the first phase of the project: a reproducible pipeline from the raw ClinVar archive to a filtered dataset, the two evaluation splits, and the initial baseline described in the proposal (logistic regression on amino-acid swap features, evaluated under a random and a gene-grouped split).

The dataset holds **199,374 missense variants in 15,272 genes** (29.6% pathogenic). The baseline reaches an **AUROC of 0.73 and an AUPRC of 0.53** (a random guess scores 0.30), with the same performance on genes never seen in training, and its probabilities are well calibrated. A diagnostic that knows only the gene, and nothing about the variant, scores an AUROC of **0.93 on the random split and exactly 0.50 on unseen genes**. This confirms the central concern of the proposal: a random split rewards recognising genes, so every later model must be judged on the gene-grouped split.

Work item (proposal section)	Status
Computing environment recorded and pinned	Done
ClinVar snapshot, filtering and leakage audit (§3, §4)	Done
Amino-acid swap features (§4)	Done
Random and gene-grouped splits, 5 seeds (§6)	Done
Initial baseline: logistic regression, both splits (§9)	Done
Calibration of the baseline (§6)	Done (preliminary)
Protein length, relative position, conservation; gradient-boosted trees (§4, §5)	Next
Level 2: ESM-2 (§5); gnomAD ablation (§4)	Next
Failure analysis, recalibration, abstention, complexity analysis (§7, §8)	Next

2. Computing environment

All results were produced on the machine below. The full record is generated by a script (`reports/environment.md`), and exact package versions are pinned in `environment-lock.yml`. A short GPU test confirmed that PyTorch runs on the GPU, which will be needed for ESM-2.

Item	Value	Item	Value
Laptop	ASUS ROG Zephyrus G14	Python	3.11.18 (conda)
Operating system	Windows 11 (build 26300)	NumPy / pandas	2.4.6 / 3.1.0
CPU	AMD Ryzen AI 9 HX 370, 12 cores / 24 threads	scikit-learn	1.9.1
RAM	31.1 GB	matplotlib / pyarrow	3.11.2 / 26.0.0
GPU	NVIDIA RTX 5070 Ti Laptop, 12 GB (driver 617.42)	PyTorch	2.11.0, CUDA 12.8 build

3. Data

Snapshot

ClinVar replaces its files every month. To keep results reproducible we downloaded the `variant_summary` file once, verified it against the checksum NCBI publishes, and recorded it. The download script verifies an existing copy instead of downloading a new one.

Source	ftp.ncbi.nlm.nih.gov/pub/clinvar/tab_delimited/variant_summary.txt.gz
Release	1 November 2026
Size / MD5	457,862,838 bytes / `d7e0d7e581e220100fc84ca64fe953ba` (matches NCBI)

Filtering

The filters of the proposal were applied in order, with three additions explained in Section 9. The number of variants after each step:

Step	Variants remaining
All rows in the snapshot	9,345,867
1. One genome version (GRCh38), one row per variant	4,636,853
2. Single-nucleotide variants	4,285,982
3. Missense (one amino acid replaced by another)	2,571,841
Start-codon changes (p.Met1…) removed	2,567,977
4. Label Pathogenic / Likely pathogenic or Benign / Likely benign	216,739
5. At least one review star	200,778
6. One row per protein change; contradictory copies dropped	199,374

Among missense variants, most labels are uncertain significance (2.17 million) or conflicting (122,000), which are removed. The combined labels "Pathogenic/Likely pathogenic" and "Benign/Likely benign" are kept in their classes; labels such as "risk factor" and "drug response" are dropped. In step 6, 1,404 variants were removed because the same amino-acid change carried a pathogenic label in one record and a benign label in another.

Final dataset

Class	Variants	Share	Gene label mix	Genes	Variants
Pathogenic (1)	59,031	29.6%	Both classes	3,281	127,656 (64.0%)
Benign (0)	140,343	70.4%	Only benign	11,791	70,187 (35.2%)
Total	199,374		Only pathogenic	200	1,531 (0.8%)

Class imbalance. About 1 pathogenic to 2.4 benign. As planned in the proposal's risk table, we report AUPRC next to AUROC; the AUPRC of a random guess equals the share of pathogenic variants, about 0.30.

Genes as a shortcut. 12,000 of the 15,272 genes contain only one class, holding 36% of all variants. For these genes, knowing the gene gives the answer, which a model can exploit on a random split. Only the 3,281 genes with both classes can be used for per-gene AUROC.

Large genes. The ten largest genes (FBN1 1,464 variants, TTN 1,242, KMT2D 1,095, SCN1A 1,053, DNAH11 1,035, BRCA2, COL4A5, LDLR, OBSCN, BRCA1) hold 4.8% of all variants, so no single gene dominates the pooled metrics.

4. Leakage audit as implemented

The rules of the proposal are enforced in code. The review status is used only to filter rows, and the last-evaluated date is stored only to define a failure-analysis slice later; neither is ever passed to a model. Submitter counts, condition names and existing predictor scores are not extracted at all. The baseline features (Section 5) describe only the amino-acid change, so they cannot carry information from the curation process.

5. Features

The initial baseline uses 16 features that describe the swap only: which amino acid was lost and which one replaced it. Nothing about the gene, the position in the protein, the review status or the date is used. Protein length, relative position and conservation will be added next.

Feature group	Features	Source
Substitution score	BLOSUM62 score of the swap	Henikoff & Henikoff (1992)
Size	Side-chain volume: original, new, change, absolute change	Zamyatnin (1972)
Hydrophobicity	Original, new, change, absolute change	Kyte & Doolittle (1982)
Charge	Original, new, change, whether it changed	Charges at physiological pH
Special residues	Cysteine, proline or glycine lost or gained	Structural roles

Every feature differs between classes in the direction biology predicts:

Feature	Benign (mean)	Pathogenic (mean)
BLOSUM62 score (low = rarely tolerated in evolution)	−0.01	−1.15
Absolute change in volume (Å³)	35.9	50.8
Absolute change in hydrophobicity	2.18	3.15
Charge changes	33%	46%
Cysteine involved	4%	14%
Glycine involved	10%	22%

The properties of the original amino acid alone barely differ between classes (for example mean original hydrophobicity is −0.63 in both): the signal lies in the change. The full table is in `reports/feature_summary.md`.

6. Evaluation design

Every variant is assigned to train (70%), validation (10%) or test (20%) in two ways, each repeated with 5 seeds. The pathogenic share is kept equal (29.6%) in every part.

- **Random split:** variants are shuffled, so a validation variant's gene is usually also in training ("new variants in familiar genes").
- **Gene split:** whole genes are assigned to one side, so no validation or test gene is ever seen in training ("unseen genes"). The script checks this property and stops if it fails.

Split (seed 0)	Part	Variants	Genes	Genes also in train
Random	train	139,560	14,291	—
	val	19,938	6,811	6,430 (94%)
	test	39,876	9,620	8,907 (93%)
Gene	train	139,466	10,704	—
	val	19,915	1,513	0
	test	39,993	3,055	0

Use of each part. Models are fitted on train, and hyperparameters are chosen by cross-validation inside train only. The results in this report are on the **validation** part. The test part has not been used and will be evaluated once, for the final results, as stated in the proposal. The split assignments are saved in the repository so both members use identical splits.

7. Initial baseline

Model. Standardised features followed by L2-regularised logistic regression. It is the interpretable reference of the model ladder.

Tuning. The regularisation strength C was searched over {0.001, 0.01, 0.1, 1, 10, 100} by 5-fold gene-grouped cross-validation inside the training part, scored by AUPRC. The tuning curve is flat: for the gene split with seed 0, cross-validated AUPRC ranges only from 0.5243 to 0.5246 across the grid. With 16 features and about 140,000 training variants the model is far from overfitting, which is also why the selected C varies between seeds without affecting performance.

No class weighting. AUROC and AUPRC depend only on how variants are ranked, which weighting barely changes, whereas weighting shifts the predicted probabilities that the calibration analysis relies on.

Gene-only reference. As a diagnostic (not part of the model ladder), each variant is predicted by the share of pathogenic variants its gene had in training, smoothed towards the overall share; unseen genes receive the overall share. It uses nothing about the variant, so its score measures how much knowing the gene alone is worth.

8. Preliminary results

Model	Split	AUROC	AUPRC	Brier ↓	Per-gene AUROC
Logistic regression	Random	0.731 ± 0.005	0.529 ± 0.005	0.178 ± 0.001	0.714 ± 0.005
Logistic regression	Gene	0.732 ± 0.005	0.530 ± 0.008	0.178 ± 0.001	0.724 ± 0.004
Gene-only reference	Random	0.931 ± 0.002	0.862 ± 0.004	0.091 ± 0.001	0.500
Gene-only reference	Gene	0.500	0.296	0.208	0.500

Validation part, mean ± standard deviation over 5 seeds. Random guess: AUROC 0.5, AUPRC 0.296. Per-gene AUROC is averaged over validation genes containing both classes (616 genes on average).

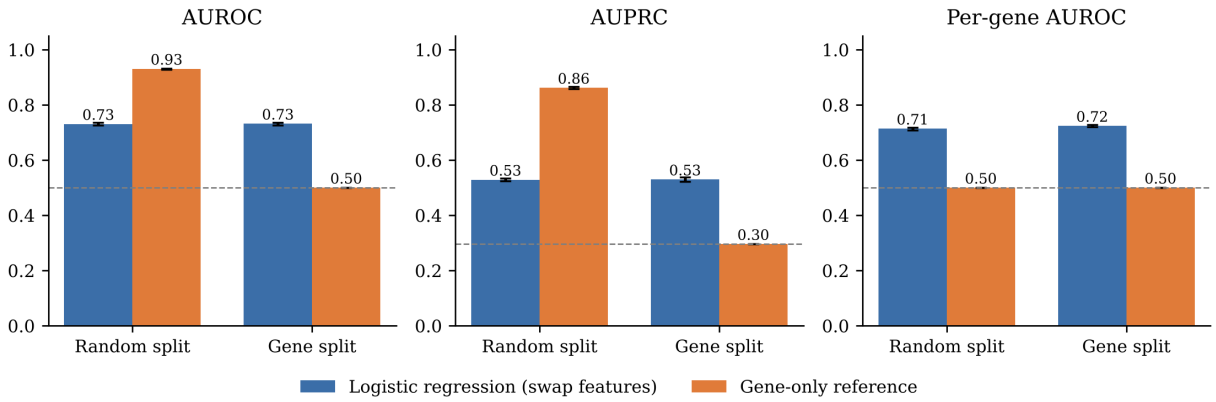


Figure 1. Initial baseline on the validation part (mean ± std over 5 seeds). Dashed lines: random guess.

Findings

1. **The swap features carry real signal.** AUPRC 0.53 against 0.30 for a random guess, from 16 numbers describing only the amino-acid change.
2. **The baseline generalises to unseen genes.** Its scores are the same on both splits (AUROC 0.731 and 0.732). Its features contain nothing that identifies the gene, so it cannot use the gene shortcut, and what it learned transfers to new genes.
3. **Gene identity alone beats the biology on a random split.** The gene-only reference scores 0.93 AUROC on the random split, far above the baseline, then falls exactly to chance on unseen genes. A random split therefore rewards recognising genes. Later features that may indirectly identify a gene, such as protein length, relative position and ESM-2 embeddings, will be judged on the gene split, and the gap between the two splits will be reported as a measure of this shortcut.
4. **Per-gene AUROC separates the two.** The baseline ranks variants within a gene (0.71–0.72), while the gene-only reference cannot (0.50). This confirms the value of the per-gene metric planned in the proposal.
5. **Results are stable across seeds** (standard deviation about 0.005).

What the model learned

Standardised coefficients (gene split, seed 0): the BLOSUM62 score is by far the strongest feature (−0.65, rarely tolerated swaps are more often pathogenic), followed by glycine involvement (+0.34), cysteine involvement (+0.24) and the volumes of the two amino acids (+0.23, +0.21). Hydrophobicity of the new amino acid and its change are negative (−0.17), meaning swaps that lose hydrophobicity lean pathogenic. Several features are strongly correlated (an original value, a new value and their difference), so individual coefficients should be read as a group rather than one by one.

Calibration

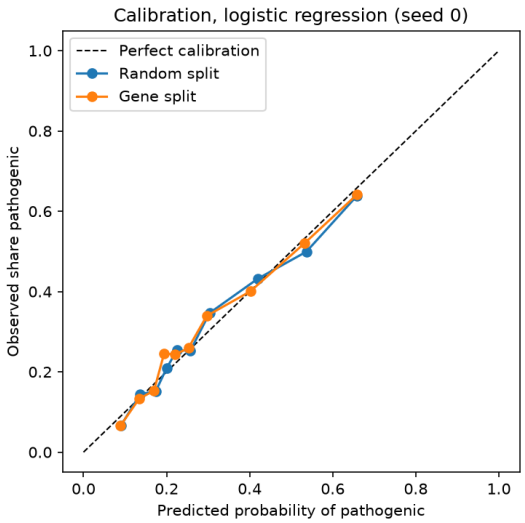


Figure 2. Reliability curves of the logistic regression on the validation part (seed 0, 10 bins with equal numbers of variants).

Both curves lie close to the diagonal: when the model predicts 30%, about 30% of those variants are pathogenic. The expected calibration error is **0.018 on both splits**, so calibration holds on unseen genes as well as on familiar ones (proposal, sub-question 3). The limitation is confidence: predictions range only from 0.07 to 0.80, so the model never states that a variant is almost certainly pathogenic or benign. A useful Level-2 model must be more confident while staying calibrated, particularly on unseen genes.

Cost

Training including tuning takes about 2 seconds on the CPU; inference takes 0.3 ms per 1,000 variants; the saved model is 1.6 KB. These figures are the reference for the complexity analysis of Level 2.

9. Changes and clarifications relative to the proposal

Item	What changed and why
Rows per variant	The file lists most variants twice, once per genome version. Only GRCh38 rows are kept.
Start-codon changes	Removed (3,864 variants). A change at p.Met1 prevents the protein from being made, a different mechanism from an amino-acid swap.
Protein-level duplicates	Different DNA changes can give the same amino-acid change with identical features. One row per protein change is kept, and contradictory copies are dropped, so the same example cannot appear on both sides of a random split.
Expected size	The proposal expected tens of thousands per class across several thousand genes. The actual dataset is 59,031 pathogenic and 140,343 benign variants in 15,272 genes.
Validation part	A fixed validation part (10%) was added to each split for preliminary results and for the recalibration planned in §7, so the test part stays untouched until the end.
Gene-only reference	Added as a diagnostic to quantify the gene shortcut directly.
Class weighting	Not used for the baseline (Section 7). AUPRC is reported throughout, as the risk table planned.

10. Reproducibility

Code, results and the trained baseline models are in the repository **github.com/M-Malek-Ht/clinvar-pathogenicity**. Every number in this report is produced by six scripts run in order, each writing its outputs to `reports/`:

```
python scripts/env_report.py        # hardware and software record
python scripts/download_clinvar.py  # download or verify the snapshot
python scripts/build_dataset.py     # filtering, with counts per step
python scripts/build_features.py    # swap features
python scripts/make_splits.py       # random and gene splits, 5 seeds
python scripts/run_baseline.py      # baseline training and evaluation
```

The notebook `notebooks/01_phase1_baseline.ipynb` walks through each step with exploration plots, the tuning curve, the results and the calibration analysis. The ClinVar file is not stored in the repository because of its size; anyone with the same snapshot (identified by the MD5 above) obtains identical numbers.

11. Next steps

Dates	Work
7–10 November	Add protein length and relative position (UniProt) and conservation (phyloP), with a check that the ClinVar reference amino acid matches the UniProt sequence. Train gradient-boosted trees as the strong Level-1 baseline. Report how much the new features change each split.
10–14 November	Level 2: ESM-2 embeddings of the wild-type and mutant sequence at the mutated position with a small classifier, starting with a small ESM-2 on the GPU; windowing for proteins longer than about 1,000 residues. Add the zero-shot ESM-2 score as a cheap comparison. Run the gnomAD allele-frequency ablation.
14–17 November	Calibration of every model on both splits, recalibration on separate validation genes, abstention option. Failure analysis on the slices of §7. Complexity analysis (time, hardware, size, interpretability).
17–20 November	Single evaluation on the test part, final report, poster, release of the final trained model.

Open risks. Mapping ClinVar positions to UniProt sequences may drop some genes (the number will be reported). ESM-2 inference on about 200,000 variants is the most expensive step; the 12 GB GPU should handle the smaller ESM-2 models, with gene subsampling as the fallback planned in the proposal.

12. References

Grimm, D. G., et al. (2015). The evaluation of tools used to predict the impact of missense variants is hindered by two types of circularity. *Human Mutation*, 36(5), 513–523.

Henikoff, S., & Henikoff, J. G. (1992). Amino acid substitution matrices from protein blocks. *PNAS*, 89(22), 10915–10919.

Karczewski, K. J., et al. (2020). The mutational constraint spectrum quantified from variation in 141,456 humans. *Nature*, 581, 434–443.

Kyte, J., & Doolittle, R. F. (1982). A simple method for displaying the hydropathic character of a protein. *Journal of Molecular Biology*, 157(1), 105–132.

Landrum, M. J., et al. (2018). ClinVar: improving access to variant interpretations and supporting evidence. *Nucleic Acids Research*, 46(D1), D1062–D1067.

Lin, Z., et al. (2023). Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637), 1123–1130.

Zamyatnin, A. A. (1972). Protein volume in solution. *Progress in Biophysics and Molecular Biology*, 24, 107–123.